### 0.0.1 Density of Rationals

**Lemma 0.0.1.** For any real numbers $a < b$, there exists a rational number $p \in \mathbb{Q}$ such that $a < p < b$.

*Proof.* If $a$ is rational, then take $N \in \mathbb{N}$ such that

$$\frac{1}{N} < b - a$$

Then, $a + \frac{1}{N}$ is a rational number that we desired. Suppose $a$ is irrational. Define a map as:

$$\lceil \cdot \rceil : \mathbb{R} \to \mathbb{Z} : x \mapsto \min\{n \in \mathbb{Z} \mid n \geq x\}$$

That is, $\lceil x \rceil$ is the smallest integer greater than or equal to $x$. And, for any $x \in \mathbb{R}$,

$$x \leq \lceil x \rceil \leq x + 1$$

By Archimedean property, there exists $N \in \mathbb{N}$ such that

$$a = \frac{Na}{N} \overset{(*)}{\leq} \frac{\lceil Na \rceil}{N} \leq \frac{Na + 1}{N} = a + \frac{1}{N} < b$$

Since $a$ is irrational, the equality of $(*)$ cannot hold. Now, $\frac{\lceil Na \rceil}{N}$ is a rational number that we desired. □

**Corollary 0.0.2.** For any real numbers $a < b$, there exists an irrational number $c \in \mathbb{R} \setminus \mathbb{Q}$ such that $a < c < b$.

*Proof.* By the Lemma, there exists a rational nubmer $p \in \mathbb{Q}$ such that

$$a < p < b$$

By Archimedean property, there exists $N \in \mathbb{N}$ such that

$$a < p < p + \frac{1}{N} < b$$

Now,

$$a < p < p + \frac{1}{2N} < p + \frac{\sqrt{2}}{2N} < p + \frac{1}{N} < b$$

□

### 0.0.2 Dirichlet function

**Theorem 0.0.3.** Define a real-valued function as:

$$f : \mathbb{R} \to \mathbb{R} : x \mapsto \begin{cases} 1 & x \in \mathbb{Q} \\ 0 & x \notin \mathbb{Q} \end{cases}$$

Then, $f$ is nowhere continuous.

*Proof.* Let $x \in \mathbb{R}$ be given. To show $f$ is not continuous at $x$ using contradiction, suppose $f$ is continuous at $x$.
By definition of continuous, there exists $\delta > 0$ such that

$$|x - t| < \delta \implies |f(x) - f(t)| < 1$$

But, the Lemma gives that there exists a rational number $p$ and an irrational number $q$ in $(x - \delta, x + \delta)$.
Now, regardless of $x$ is rational or irrational, we can choose $t = p$ or $t = q$ to make $|f(x) - f(t)| = 1$, contradiction. □